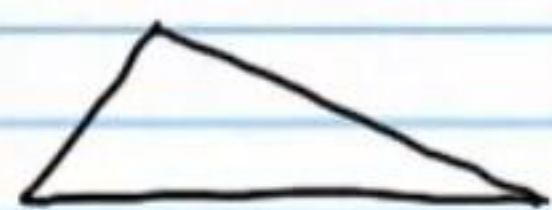


An Introduction to Triangles: The Pythagorean Theorem and the Triangle Sum Theorem

Classifying Triangles by Sides

Scalene Triangle



Isosceles Triangle

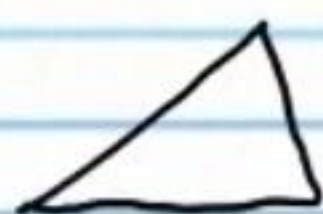


Equilateral Triangle

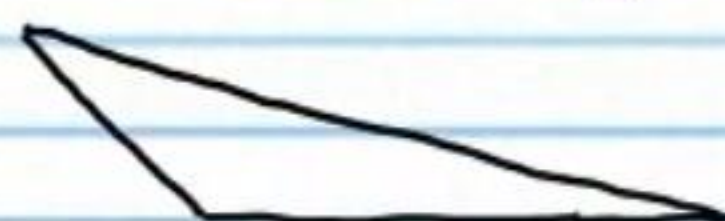


Angles

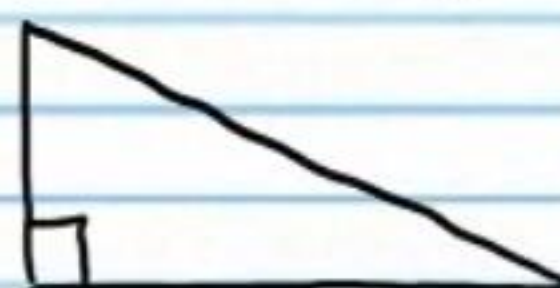
Acute Triangle



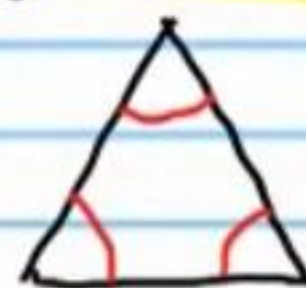
Obtuse Triangle



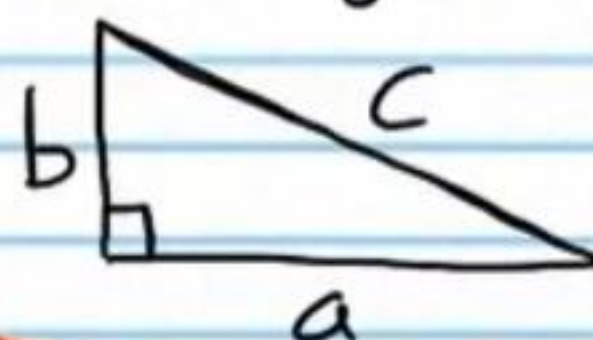
Right Triangle



Equiangular Triangle

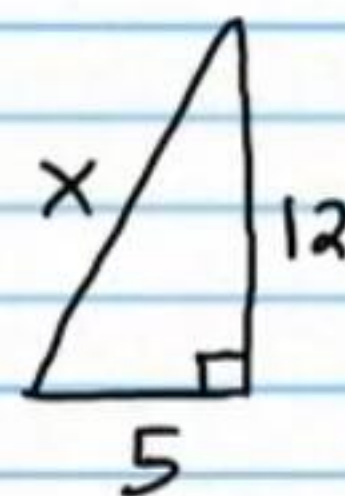


The Pythagorean Theorem: If a , b , and c are the legs and hypotenuse of a right triangle, respectively, then $a^2 + b^2 = c^2$.



Find the missing side of each right triangle.

①



$$a^2 + b^2 = c^2$$

$$5^2 + 12^2 = x^2$$

$$5 \cdot 5 + 12 \cdot 12 = x^2$$

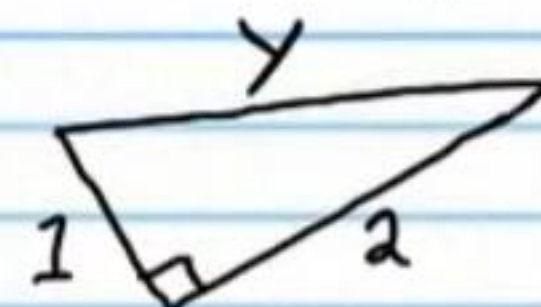
$$25 + 144 = x^2$$

$$169 = x^2$$

$$\sqrt{169} = \sqrt{x^2}$$

$$13 = x$$

②



$$a^2 + b^2 = c^2$$

$$1^2 + 2^2 = y^2$$

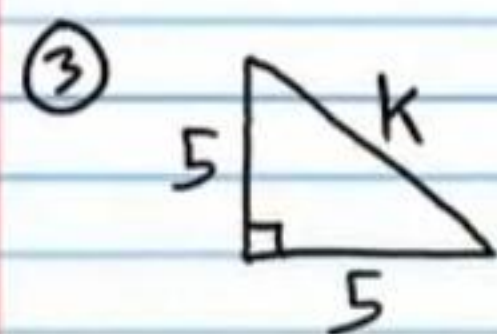
$$1 + 4 = y^2$$

$$5 = y^2$$

$$\sqrt{5} = \sqrt{y^2}$$

$$\sqrt{5} = y$$

$$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$$



$$a^2 + b^2 = c^2$$

$$5^2 + 5^2 = k^2$$

$$25 + 25 = k^2$$

$$50 = k^2$$

$$\sqrt{50} = \sqrt{k^2}$$

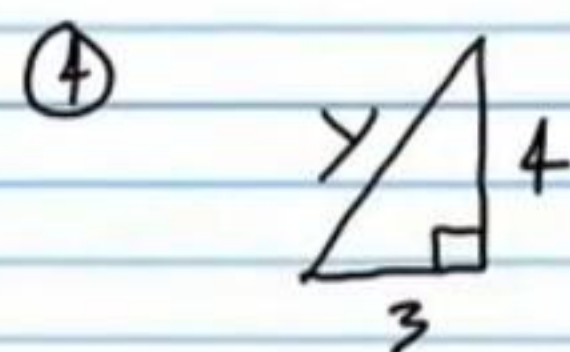
$$\sqrt{50} = k$$

$$\sqrt{25 \cdot 2} = k$$

$$\sqrt{25} \cdot \sqrt{2} = k$$

$$5 \cdot \sqrt{2} = k$$

$$\boxed{5\sqrt{2} = k}$$



$$a^2 + b^2 = c^2$$

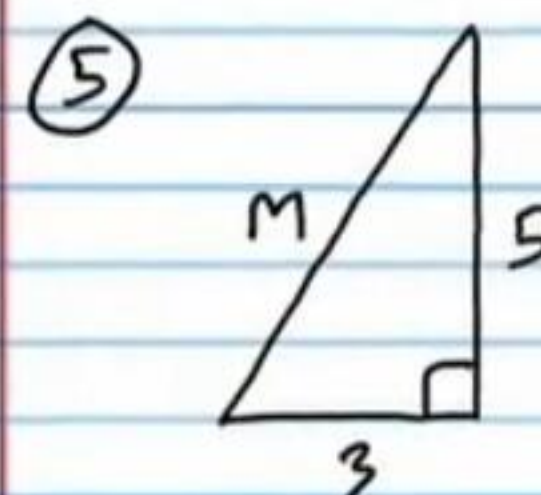
$$4^2 + 3^2 = y^2$$

$$16 + 9 = y^2$$

$$25 = y^2$$

$$\sqrt{25} = \sqrt{y^2}$$

$$\boxed{5 = y}$$



$$a^2 + b^2 = c^2$$

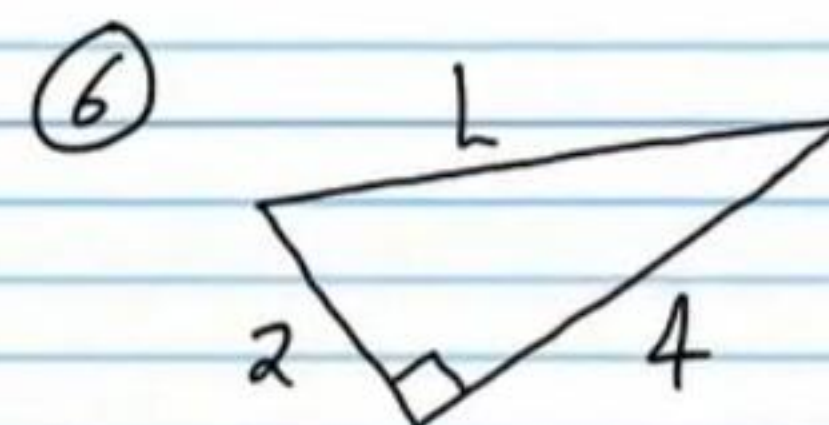
$$3^2 + 5^2 = m^2$$

$$9 + 25 = m^2$$

$$34 = m^2$$

$$\sqrt{34} = \sqrt{m^2}$$

$$\boxed{\sqrt{34} = m}$$



$$a^2 + b^2 = c^2$$

$$2^2 + 4^2 = L^2$$

$$4 + 16 = L^2$$

$$20 = L^2$$

$$\sqrt{20} = \sqrt{L^2}$$

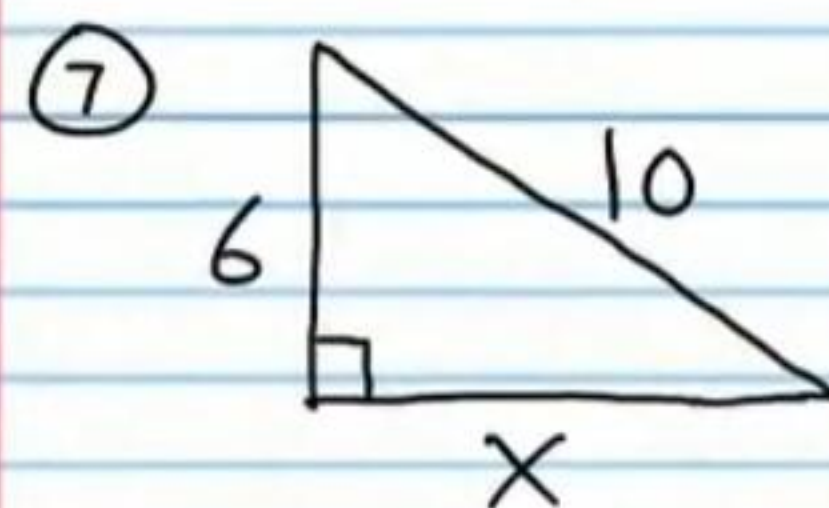
$$\sqrt{20} = L$$

$$\sqrt{4 \cdot 5} = L$$

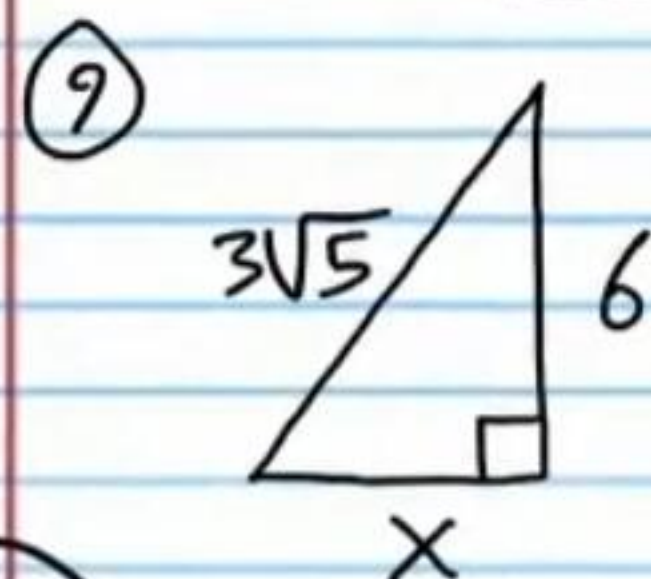
$$\sqrt{4} \cdot \sqrt{5} = L$$

$$2 \cdot \sqrt{5} = L$$

$$\boxed{2\sqrt{5} = L}$$

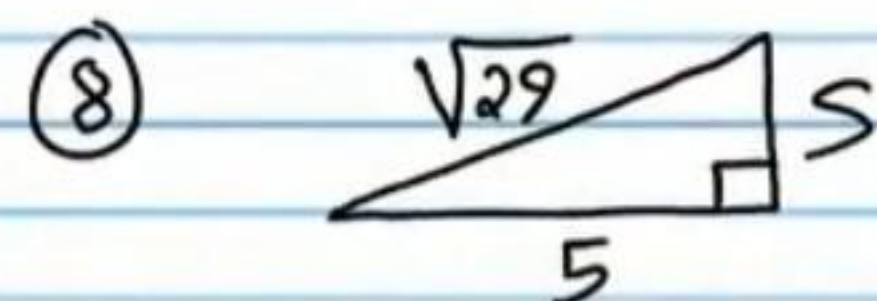


$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 6^2 + x^2 &= 10^2 \\
 36 + x^2 &= 100 \\
 x^2 &= 64 \\
 \sqrt{x^2} &= \sqrt{64} \\
 \boxed{x = 8}
 \end{aligned}$$

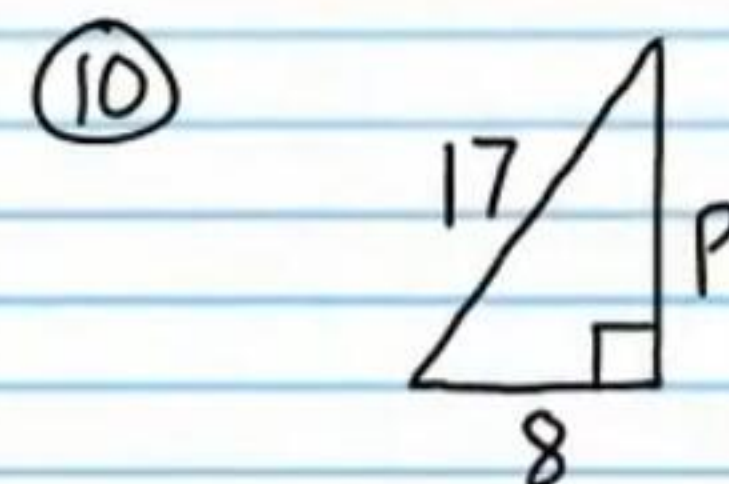


$x^2 = 9$
 $\sqrt{x^2} = \sqrt{9}$
 $\boxed{x = 3}$

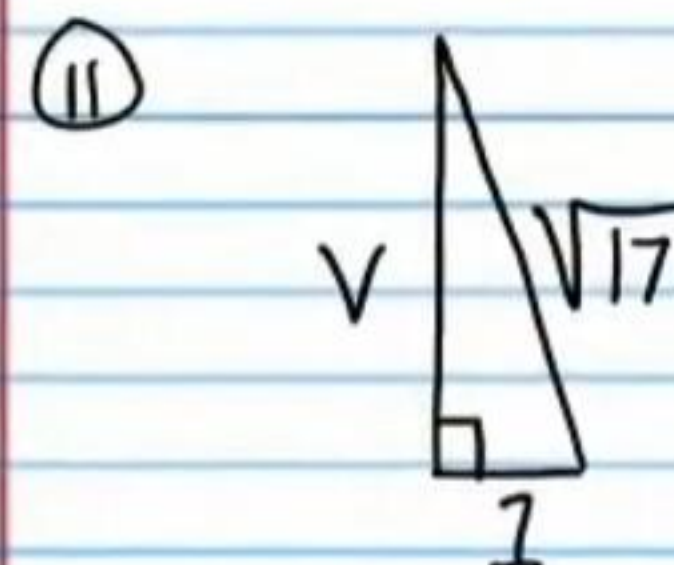
$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 x^2 + 6^2 &= (3\sqrt{5})^2 \\
 x^2 + 36 &= (3\sqrt{5})(3\sqrt{5}) \\
 x^2 + 36 &= 9 \cdot 5 \\
 x^2 + 36 &= 45
 \end{aligned}$$



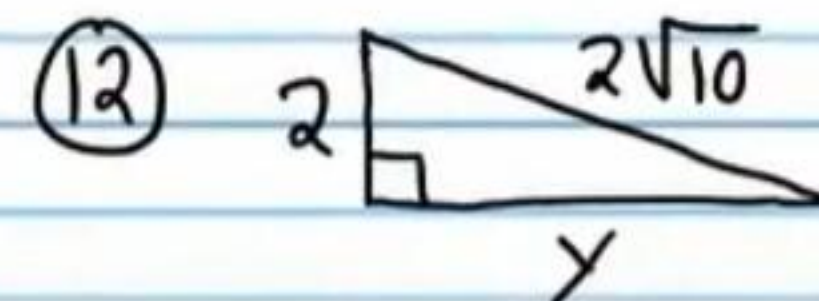
$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 5^2 + s^2 &= (\sqrt{29})^2 \\
 25 + s^2 &= 29 \\
 s^2 &= 4 \\
 \sqrt{s^2} &= \sqrt{4} \\
 \boxed{s = 2}
 \end{aligned}$$



$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 8^2 + p^2 &= 17^2 \\
 64 + p^2 &= 289 \\
 p^2 &= 225 \\
 \sqrt{p^2} &= \sqrt{225} \\
 \boxed{p = 15}
 \end{aligned}$$

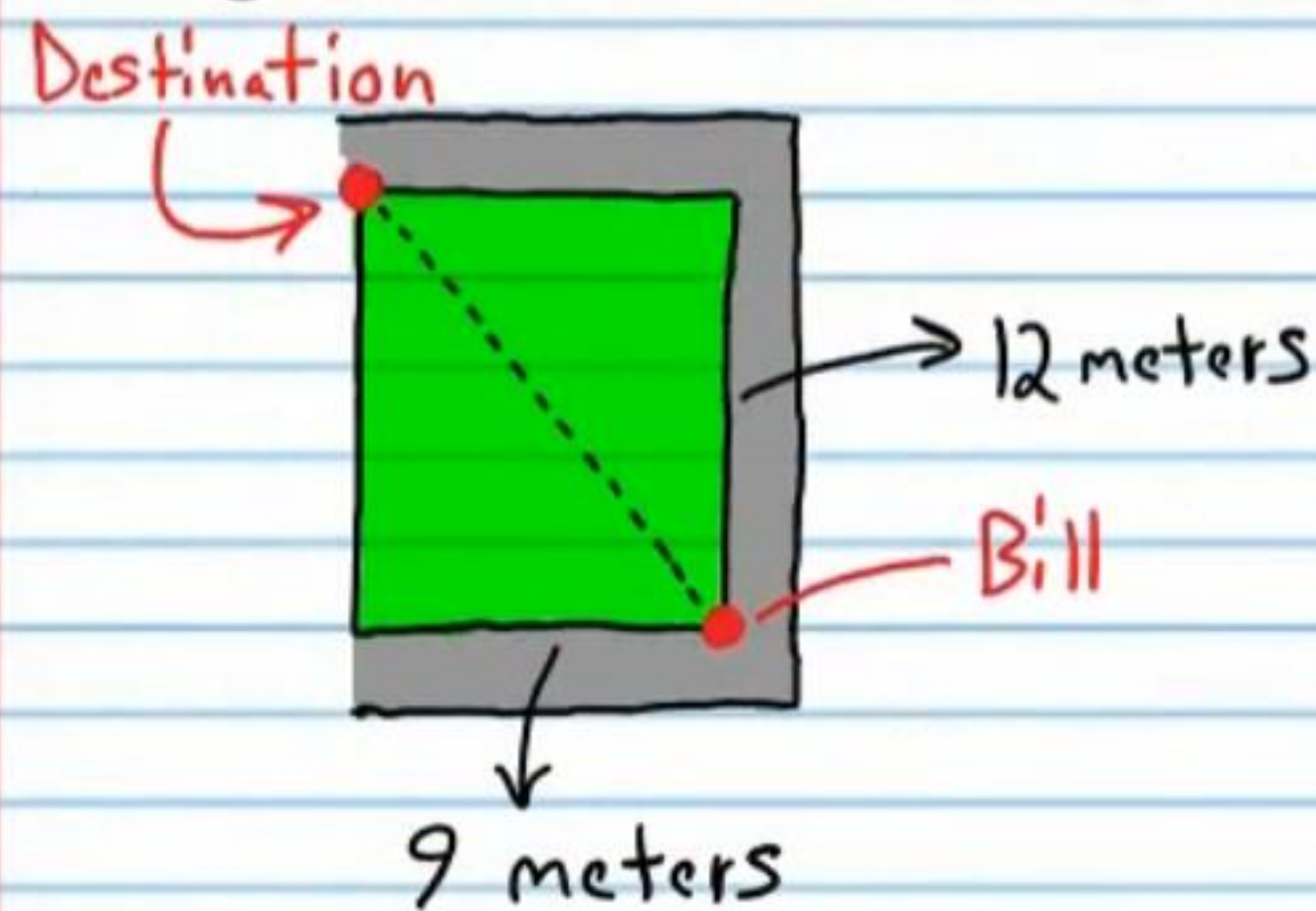


$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 v^2 + 1^2 &= (\sqrt{17})^2 \\
 v^2 + 1 &= 17 \\
 v^2 &= 16 \\
 \sqrt{v^2} &= \sqrt{16} \\
 \boxed{v = 4}
 \end{aligned}$$

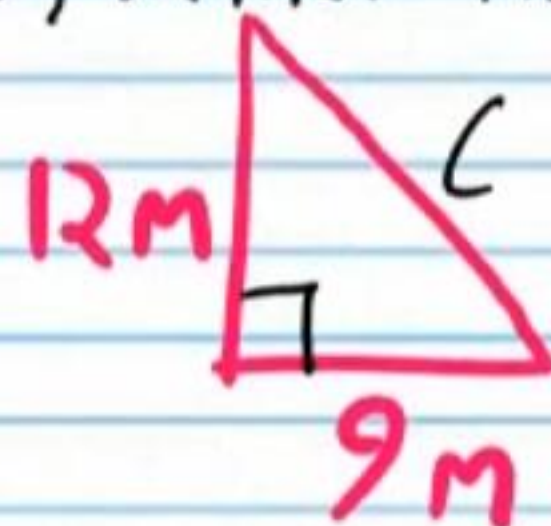


$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 2^2 + y^2 &= (2\sqrt{10})^2 \\
 4 + y^2 &= (2\sqrt{10})(2\sqrt{10}) \\
 4 + y^2 &= 4 \cdot \sqrt{100} \\
 4 + y^2 &= 4 \cdot 10 \\
 4 + y^2 &= 40 \\
 y^2 &= 36 \\
 \sqrt{y^2} &= \sqrt{36} \\
 \boxed{y = 6}
 \end{aligned}$$

⑬ How much shorter will Bill's path be if he walks across the grass, rather than using the walkway?

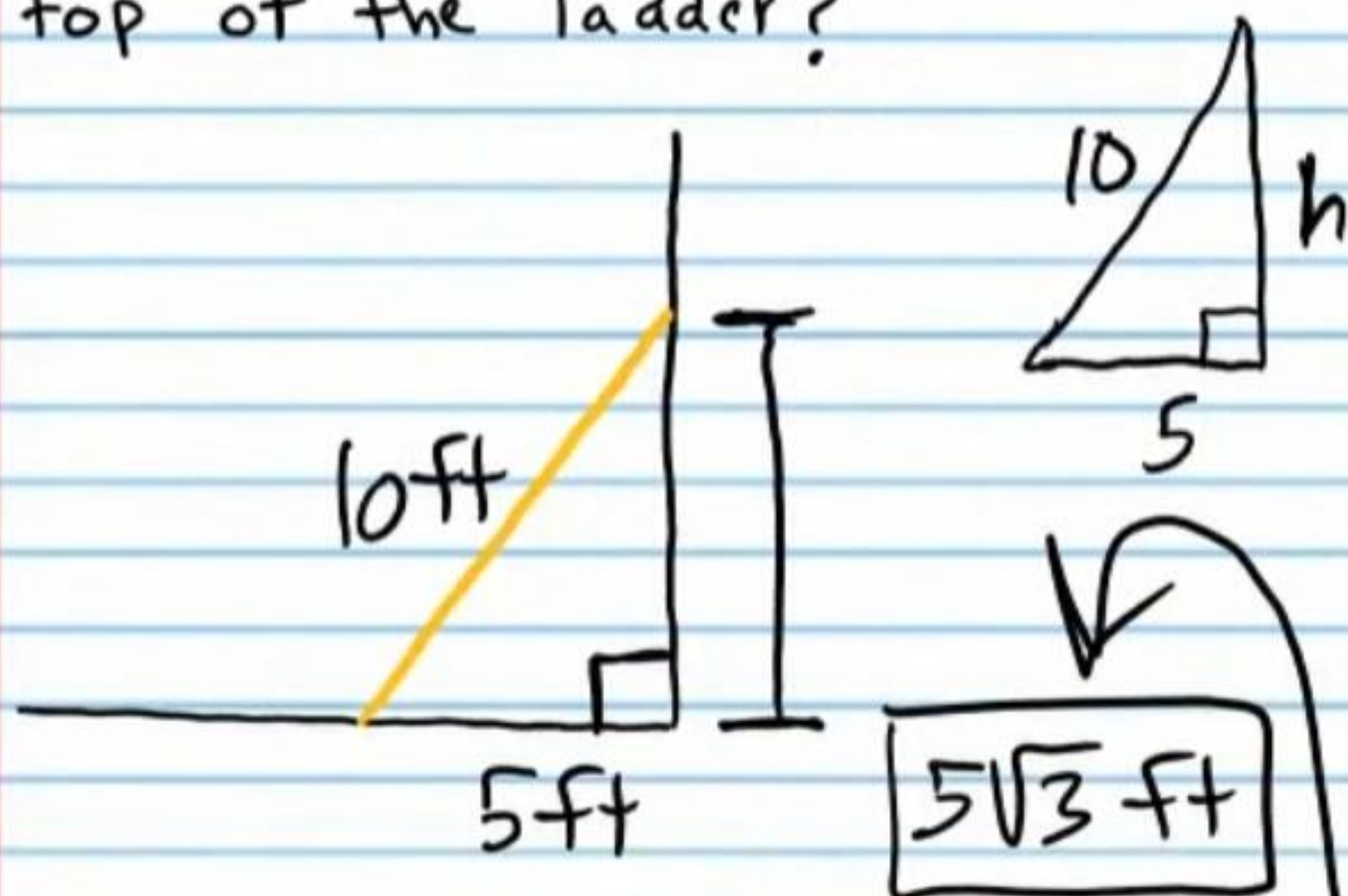


$$12 + 9 - 15 = \boxed{6 \text{ m}}$$



$$\begin{aligned} a^2 + b^2 &= c^2 \\ 12^2 + 9^2 &= c^2 \\ 144 + 81 &= c^2 \\ 225 &= c^2 \\ 15 &= c \end{aligned}$$

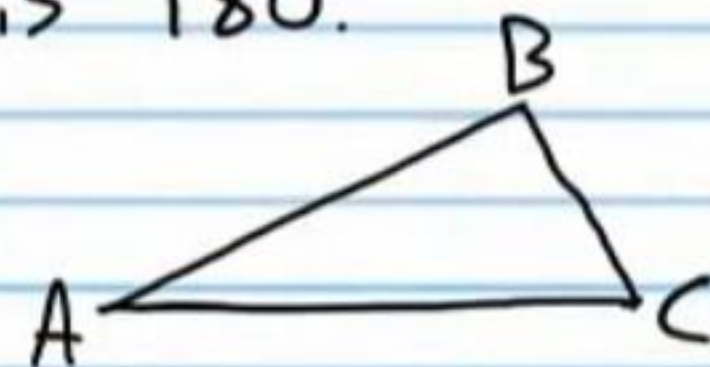
⑭ A 10 foot ladder is resting against a wall. If the bottom of the ladder is five feet from the wall, what is the height of the top of the ladder?



$$\boxed{5\sqrt{3} \text{ ft}}$$

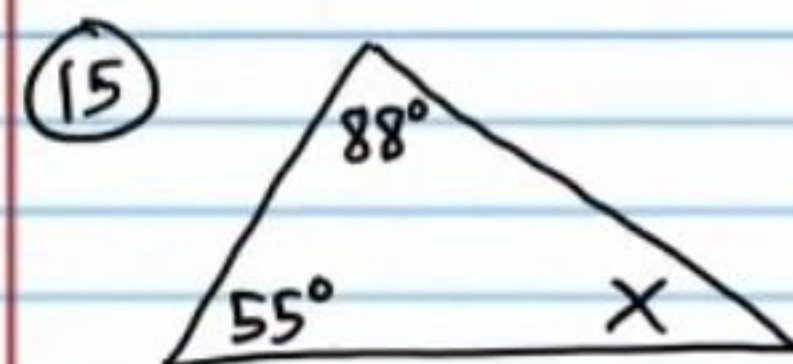
$$\begin{aligned} a^2 + b^2 &= c^2 \\ 5^2 + h^2 &= 10^2 \\ 25 + h^2 &= 100 \\ h^2 &= 75 \\ h &= \sqrt{75} \\ h &= \sqrt{25 \cdot 3} \\ h &= 5\sqrt{3} \end{aligned}$$

The Triangle Sum Theorem: The sum of the measures of the interior angles of any triangle is 180° .



$$m\angle A + m\angle B + m\angle C = 180^\circ$$

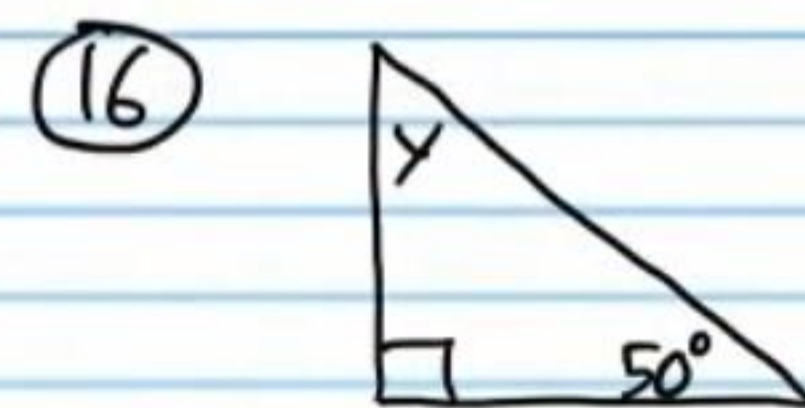
Find the value of each variable.



$$88 + 55 + x = 180$$

$$143 + x = 180$$

$$\boxed{x = 37^\circ}$$

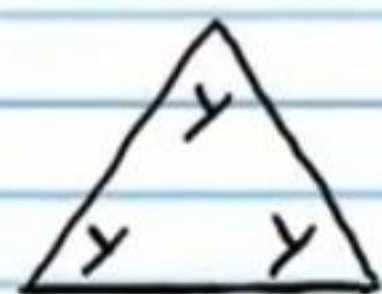


$$y + 90 + 50 = 180$$

$$y + 140 = 180$$

$$\boxed{y = 40^\circ}$$

(17)

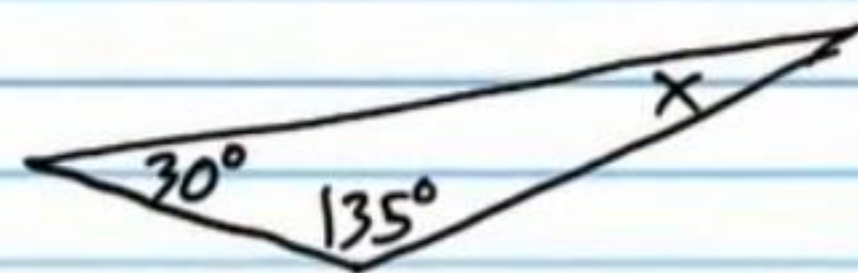


$$y + y + y = 180$$

$$3y = 180$$

$$\boxed{y = 60^\circ}$$

(18)

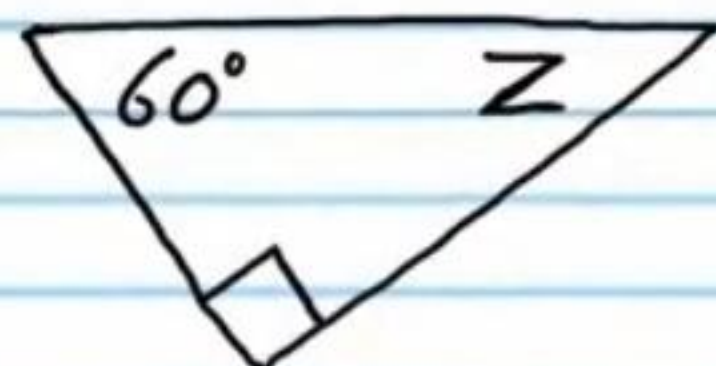


$$30 + 135 + x = 180$$

$$165 + x = 180$$

$$\boxed{x = 15^\circ}$$

(19)

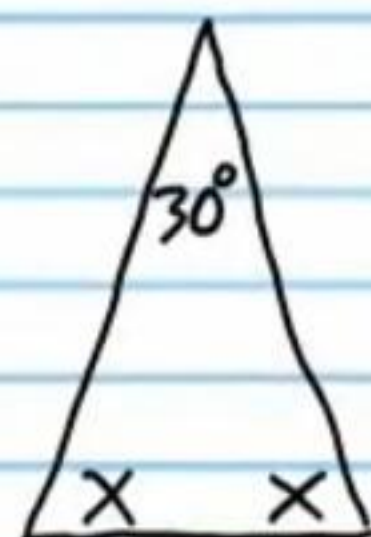


$$60 + 90 + z = 180$$

$$150 + z = 180$$

$$\boxed{z = 30^\circ}$$

(20)



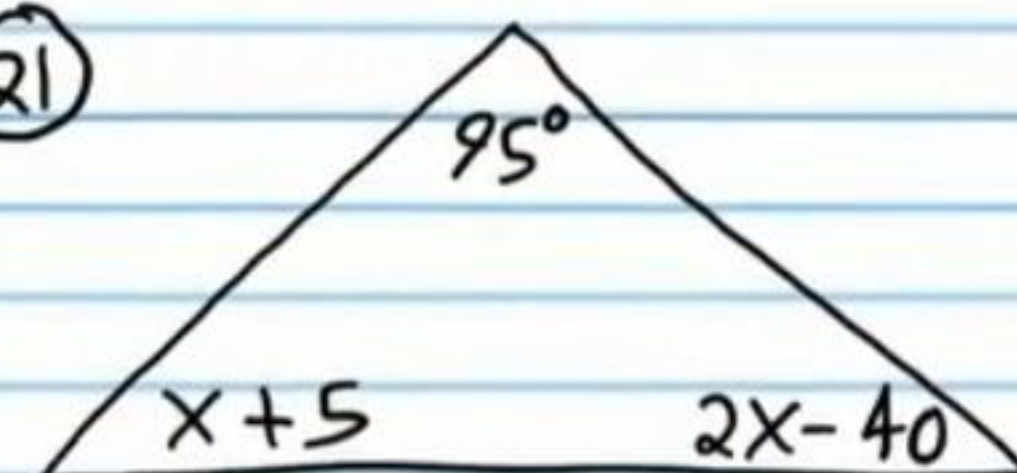
$$x + x + 30 = 180$$

$$2x + 30 = 180$$

$$2x = 150$$

$$\boxed{x = 75^\circ}$$

(21)



$$95 + (x+5) + (2x-40) = 180$$

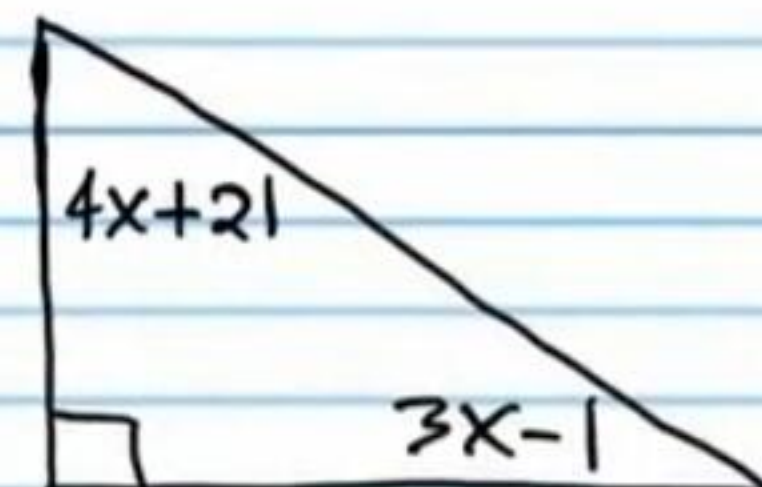
$$95 + x + 5 + 2x - 40 = 180$$

$$3x + 60 = 180$$

$$3x = 120$$

$$\boxed{x = 40}$$

(22)



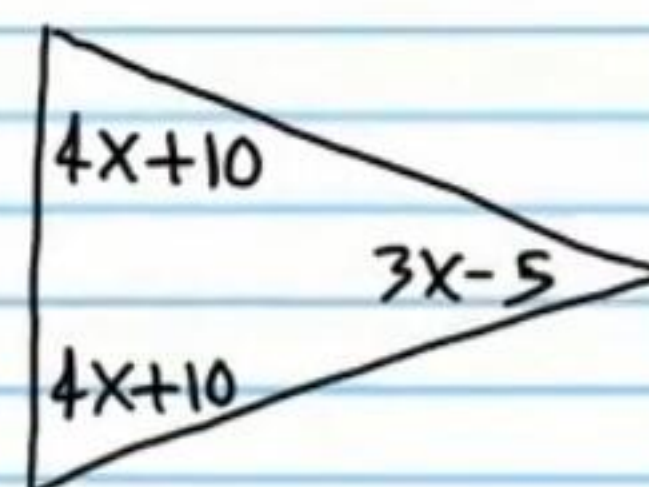
$$4x + 21 + 90 + 3x - 1 = 180$$

$$7x + 110 = 180$$

$$7x = 70$$

$$\boxed{x = 10}$$

(23)



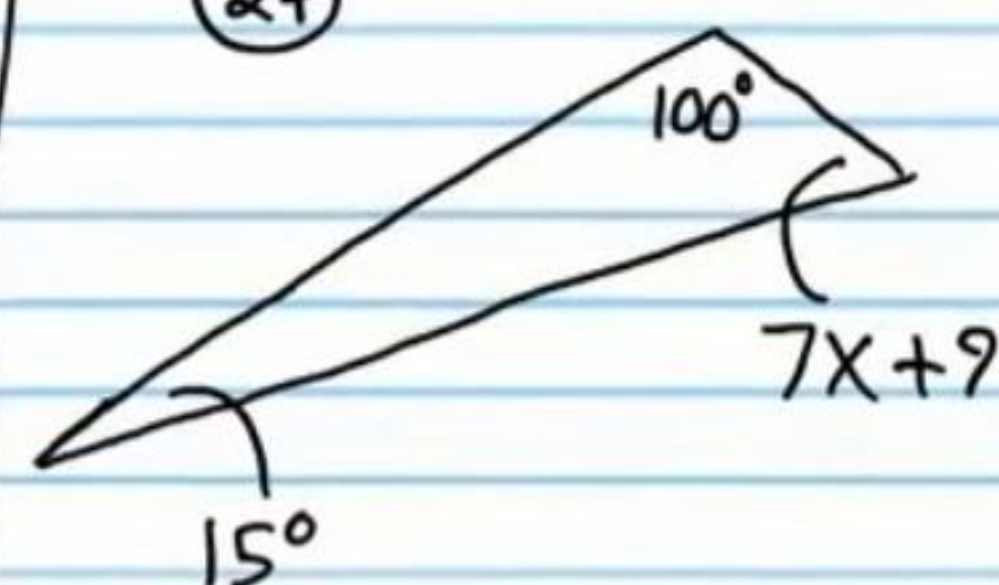
$$4x + 10 + 4x + 10 + 3x - 5 = 180$$

$$11x + 15 = 180$$

$$11x = 165$$

$$\boxed{x = 15}$$

(24)



$$100 + 15 + 7x + 9 = 180$$

$$124 + 7x = 180$$

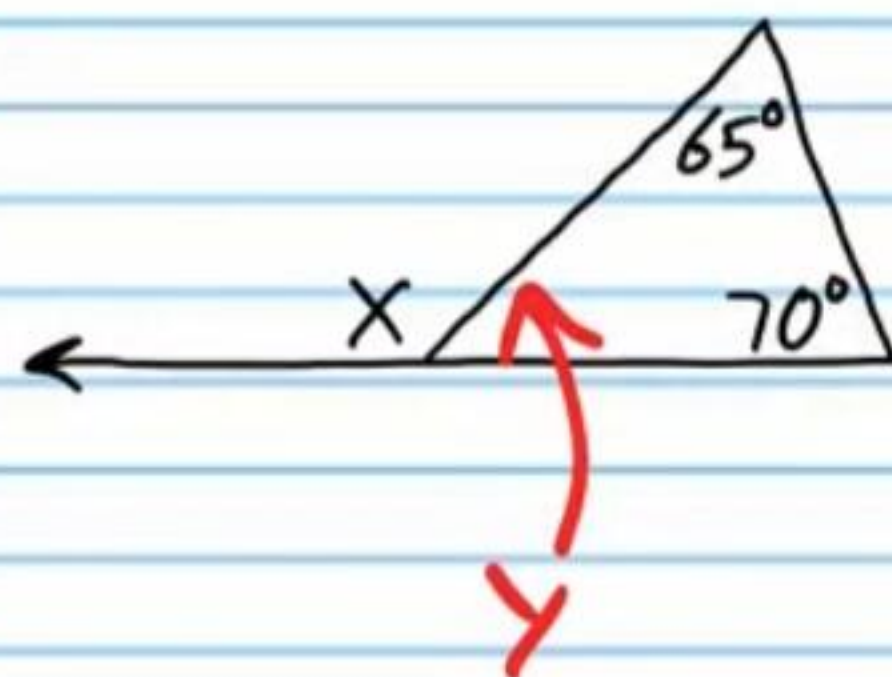
$$7x = 56$$

$$\boxed{x = 8}$$

Challenge Problems

Find x.

(25)



$$\begin{aligned} \text{I) } 65 + y + 70 &= 180 \\ y + 135 &= 180 \end{aligned}$$

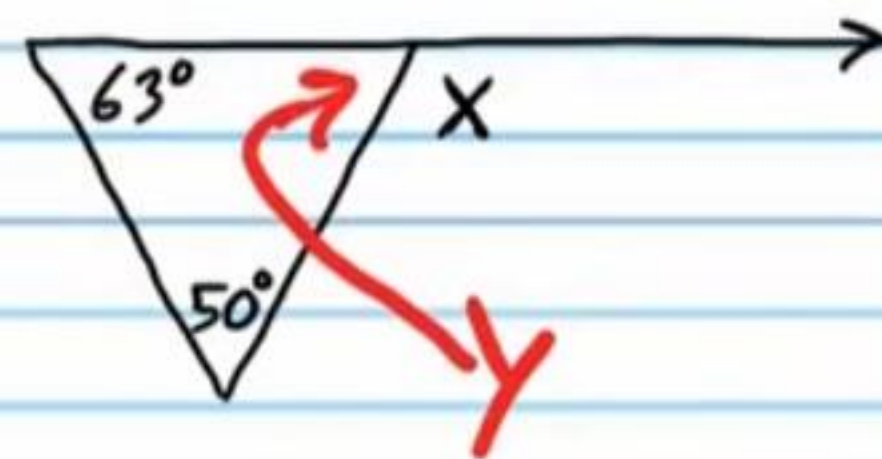
$$y = 45$$

$$\text{II) } x + y = 180$$

$$x + 45 = 180$$

$$\boxed{x = 135^\circ}$$

(26)



$$\text{I) } 63 + 50 + y = 180$$

$$113 + y = 180$$

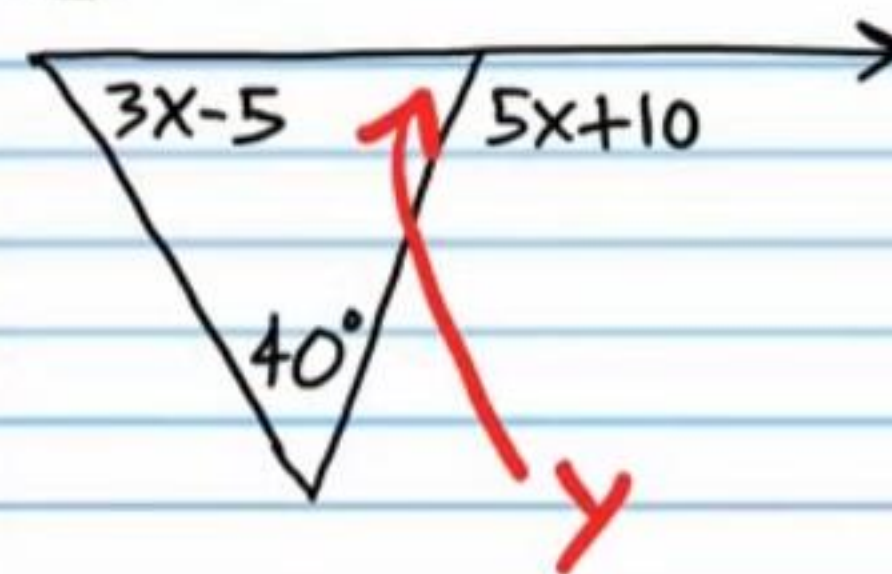
$$y = 67$$

$$\text{II) } x + y = 180$$

$$x + 67 = 180$$

$$\boxed{x = 113^\circ}$$

(27)



$$\begin{cases} 3x - 5 + y + 40 = 180 \end{cases}$$

$$\begin{cases} 5x + 10 + y = 180 \end{cases}$$

$$\begin{cases} 3x + y + 35 = 180 \end{cases}$$

$$\begin{cases} 5x + 10 + y = 180 \end{cases}$$

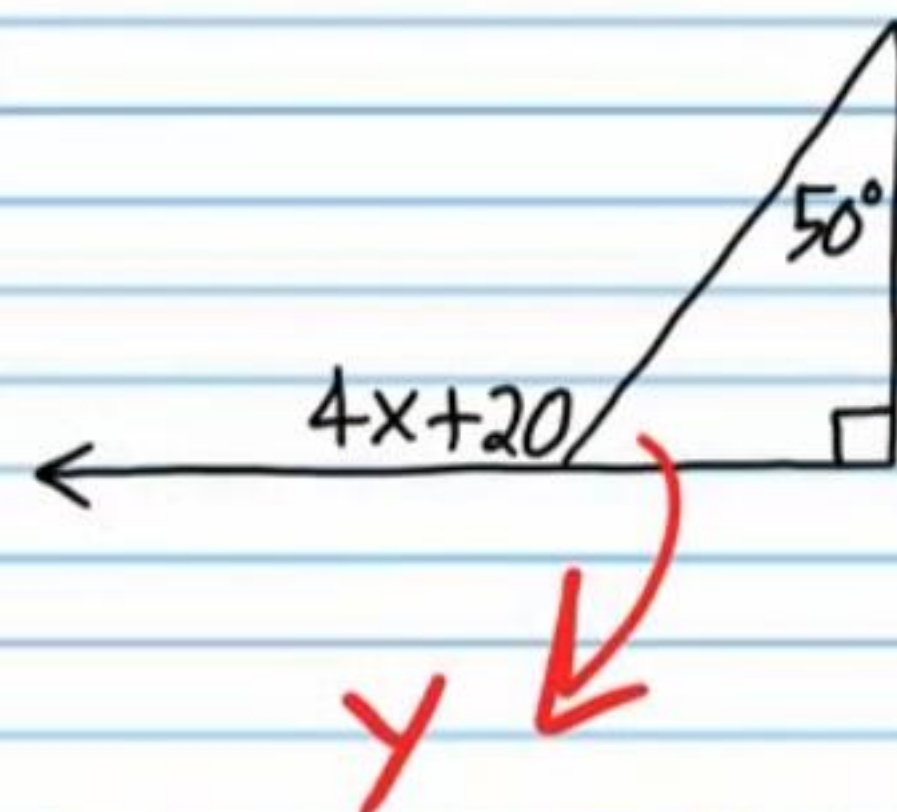
$$\begin{cases} 3x + y = 145 \end{cases}$$

$$\begin{cases} 5x + y = 170 \end{cases}$$

$$\begin{array}{r} -2x \quad = -25 \end{array}$$

$$\boxed{x = 12.5}$$

(28)



$$\begin{cases} 50 + 90 + y = 180 \end{cases}$$

$$\begin{cases} 4x + 20 + y = 180 \end{cases}$$

$$\begin{cases} 140 + y = 180 \end{cases}$$

$$\begin{cases} 4x + y = 160 \end{cases}$$

$$\begin{cases} y = 40 \end{cases}$$

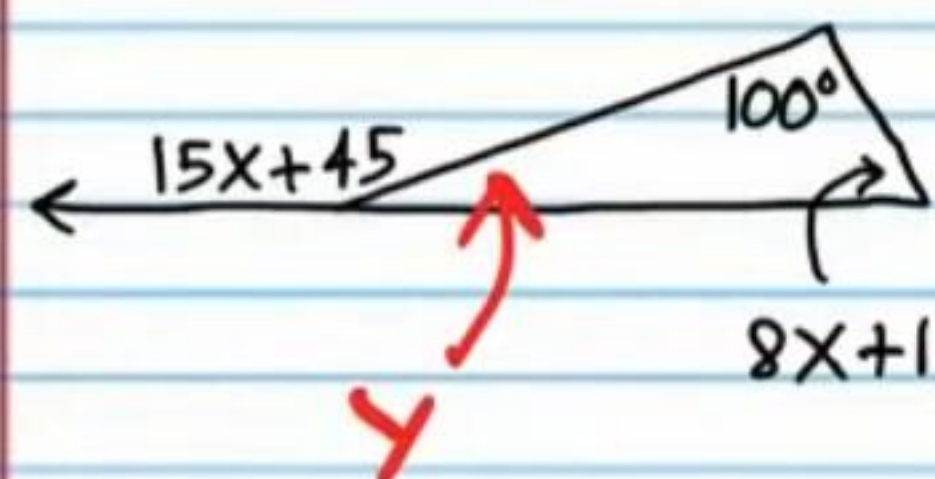
$$\begin{cases} 4x + y = 160 \end{cases}$$

$$4x + 40 = 160$$

$$4x = 120$$

$$\boxed{x = 30}$$

(29)



$$\begin{cases} y + 100 + 8x + 1 = 180 \end{cases}$$

$$\begin{cases} 15x + 45 + y = 180 \end{cases}$$

$$\begin{cases} 8x + y + 101 = 180 \end{cases}$$

$$\begin{cases} 15x + y + 45 = 180 \end{cases}$$

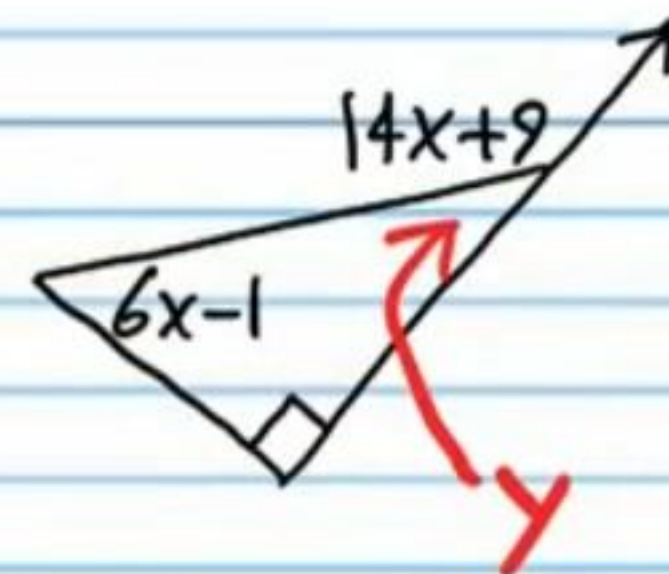
$$\begin{cases} 8x + y = 79 \end{cases}$$

$$\begin{cases} 15x + y = 135 \end{cases}$$

$$\begin{array}{r} -7x \\ \hline = -56 \end{array}$$

$$\boxed{x = 8}$$

(30)



$$\begin{cases} 6x - 1 + 90 + y = 180 \end{cases}$$

$$\begin{cases} 14x + 9 + y = 180 \end{cases}$$

$$\begin{cases} 6x + y = 91 \end{cases}$$

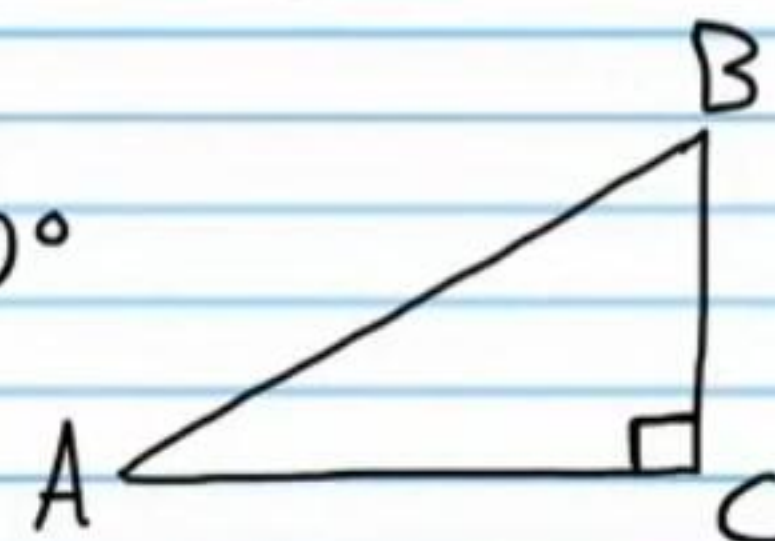
$$\begin{cases} 14x + y = 171 \end{cases}$$

$$\begin{array}{r} -8x \\ \hline = -80 \end{array}$$

$$\boxed{x = 10}$$

Show that the following theorems are true with two-column proofs.

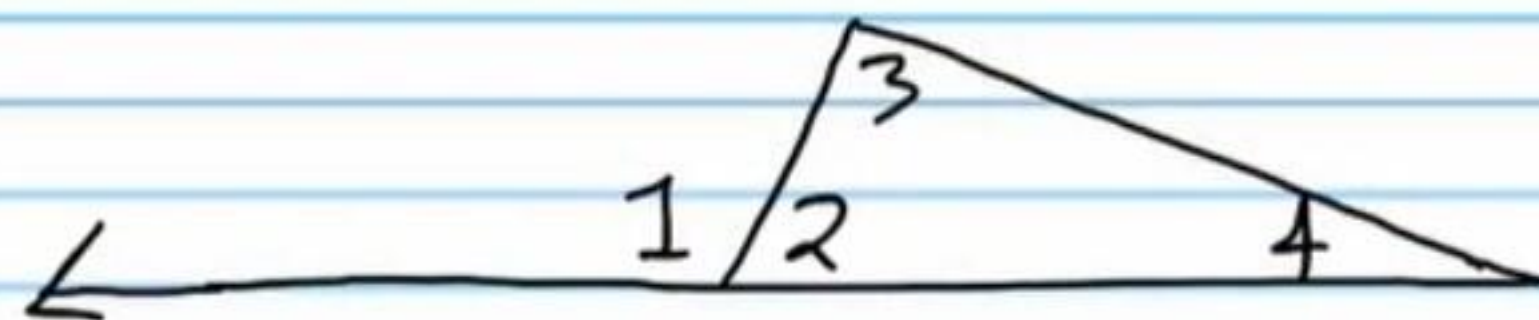
③ Right Triangle Complements Theorem: The two acute angles in a right triangle are complementary.



$$m\angle A + m\angle B = 90^\circ$$

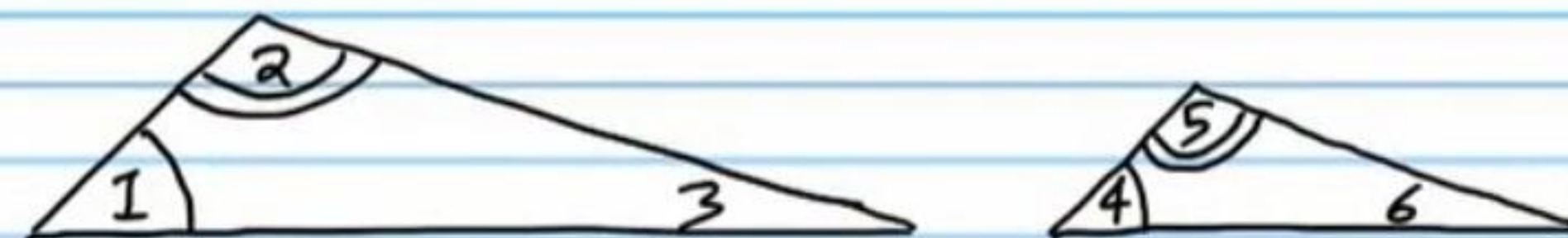
Statement	Reason
I) $\angle C$ is a rt. $\angle$.	Given.
II) $m\angle C = 90^\circ$	Def. of rt. $\angle$.
III) $m\angle A + m\angle B + m\angle C = 180^\circ$	Δ Sum. Thm.
IV) $m\angle A + m\angle B + 90^\circ = 180^\circ$	Subs. Prop of =.
V) $m\angle A + m\angle B = 90$	Subt. Prop of =.
VI) $\angle A$ & $\angle B$ are comp $\angle$'s.	Def. of comp $\angle$'s.

③② Exterior Angle Theorem: The measure of an exterior angle of a triangle is equal to the sum of the measures of the two nonadjacent interior angles.



Statement	Reason
I) Figure	Given.
II) $m\angle 2 + m\angle 3 + m\angle 4 = 180^\circ$	Δ Sum Thm.
III) $\angle 1$ & $\angle 2$ form a linear pair	Given (diagram)
IV) $\angle 1$ & $\angle 2$ are supp. $\angle$'s	Linear Pair Post.
V) $m\angle 1 + m\angle 2 = 180^\circ$	Def. of supp $\angle$'s.
VI) $m\angle 2 + m\angle 3 + m\angle 4 = m\angle 1 + m\angle 2$	Trans. Prop of =.
VII) $m\angle 3 + m\angle 4 = m\angle 1$	Subt. Prop of =.

③③ Third Angles Theorem: If the measures of two angles in one triangle are equal to the measures of two angles in another triangle, then the measures of the third angles in the triangles are equal.



Statement	Reason
I) $m\angle 2 = m\angle 5$ & $m\angle 1 = m\angle 4$	Given.
II) $m\angle 2 + m\angle 1 + m\angle 3 = 180$ and $m\angle 4 + m\angle 5 + m\angle 6 = 180$	Δ Sum Thm.
III) $m\angle 2 + m\angle 1 + m\angle 3 = m\angle 4 + m\angle 5 + m\angle 6$	Trans. Prop of =
IV) $m\angle 5 + m\angle 4 + m\angle 3 = m\angle 4 + m\angle 5 + m\angle 6$	Subs. Prop of =.
V) $m\angle 3 = m\angle 6$	Subt. Prop of =.